Performance Testing

Date	30 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Performance Testing

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman and Manual Load Testing
Type of Testing	Load Testing and Stress Testing
Target Module / API	Flood Prediction Endpoint and Flask Web Application
Test Environment	Local System (Flask Server)
Test Date	30 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user prediction request	1	30	Prediction generated successfully.
2	Multiple prediction requests	10	60	System handles concurrent requests without errors.
3	Peak load simulation	25	60	Application remains responsive under increased load.
4	Continuous prediction requests	15	120	Stable performance without failures.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Predictions generated quickly.
2	Response Time (Max)	< 5 seconds	3.4 seconds	Pass	Maximum delay within limits.
3	Throughput (Req/sec)		7 req/sec	Pass	Efficient request handling.
4	Error Rate	< 1%	0%	Pass	No errors observed.
5	CPU Utilization	< 80%	45%	Pass	CPU usage remained stable.
6	Memory Utilization	< 80%	52%	Pass	Memory usage within limits.

Step 4: Observations & Analysis

The flood prediction system demonstrated stable performance under both normal and peak load conditions.The application maintained acceptable response times and handledprediction requests efficiently.No critical errors were observed during testing.CPU and memory utilization remained within acceptable limits.

Step 5: Screenshots / Evidence

Flood Prediction Result generated by the Flask web application